Lab 8 Activity

PSYC 6802: Intro to Psychological Statistics

1

Load in the `Ornstein` data from the `carData` package, along with the `tidyverse` package. Read the help page for the `Ornstein` data to better understand these data.

```
library(tidyverse)
data("Ornstein", package = "carData")
```

For today, we will use `interlocks` as the DV, and `sector` and `nation` as IVs. However, we are only interested in whether the nation is Canada or not, so we want to combine the `OTH`, `UK`, and `US` factor levels. One way to accomplish this is to run the following code. (Then use `canada` as an IV instead of `nation`).

```
Ornstein <- Ornstein %>%
  mutate(canada = factor(nation == "CAN"))
```

Finally, we will omit data from the "BNK", "CON", and "HLD" levels of `sector` because they have relatively few observations.

```
Ornstein <- Ornstein %>%
  filter(!sector %in% c("BNK", "CON", "HLD"), .preserve = FALSE) %>%
  mutate(sector = factor(sector))
```

2

Conduct graphical and numerical exploratory analyses on these data. How do these analyses inform how you set up the ANOVA and what results you expect to find?

3

Conduct a two-way ANOVA on these data using options informed by the previous step, and calculate effect sizes. What do you find?

4

Based on your results in the previous part, calculate either simple main effects or ordinary main effects. What do you conclude?